Fórmulas Intervalos de confianza
Profesora: Diana C. Moreno Ch.

| $\theta$ | **Intervalo de confianza** | **IC** |
|---|---|---|
| $\mu$ | $\bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$ | Muestras grandes $\sigma$ conocido |
| $\mu$ | $\bar{X} - t_{(\alpha/2;gl)}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{(1-\alpha/2;gl)}\frac{s}{\sqrt{n}}$ | Muestras pequeñas |
| $\mu_1 - \mu_2$ | $(\bar{x_1} - \bar{x_2}) - Z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \leq \mu_1 - \mu_2 \leq (\bar{x_1} - \bar{x_2}) + Z_{(1-\alpha/2)}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ | Muestras grandes |
| $\mu_1 - \mu_2$ | $\bar{x_1} - \bar{x_2} - t\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)} \leq \mu_1 - \mu_2 \leq \bar{x_1} - \bar{x_2} + t\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ $s^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$ | Muestras pequeñas e independientes varianzas desconocidas pero $\sigma_1^2 = \sigma_2^2$ |
| $\mu_1 - \mu_2$ | $(\bar{y_1} - \bar{y_2}) - t_{\alpha/2;gl}\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)} \leq \mu_1 - \mu_2 \leq (\bar{y_1} - \bar{y_2}) + t_{\alpha/2;gl}\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}$ $gl = \frac{[s_1^2/n_1 + s_2^2/n_2]^2}{[(s_1^2/n_1)^2/(n_1-1)] + [(s_2^2/n_2)^2/(n_2-1)]}$ | Muestras pequeñas e independientes varianzas desconocidas y $\sigma_1^2 \neq \sigma_2^2$ |
| $\mu_1 - \mu_2$ | $\bar{d} - t\left(\frac{s_d}{\sqrt{n}}\right) \leq (\mu_1 - \mu_2) \leq \bar{d} + t\left(\frac{s_d}{\sqrt{n}}\right)$ $\bar{d} = \frac{\sum d_i}{n} \qquad s_d = \sqrt{\frac{\sum d_i^2 - \frac{\sum d_i^2}{n}}{n-1}}$ | Muestras pequeñas dependientes |
| $p$ | $\hat{p} - Z_{(\alpha/2)}\sqrt{\frac{\hat{p}\hat{q}}{n}} \leq p \leq \hat{p} + Z_{1-\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$ | Proporción |
| $p_1 - p_2$ | $(\hat{p_1} - \hat{p_2}) - Z_{\alpha/2}\sqrt{\frac{\hat{p_1}\hat{q_1}}{n_1} + \frac{\hat{p_2}\hat{q_2}}{n_2}} \leq p_1 - p_2 \leq (\hat{p_1} - \hat{p_2}) + Z_{1-\alpha/2}\sqrt{\frac{\hat{p_1}\hat{q_1}}{n_1} + \frac{\hat{p_2}\hat{q_2}}{n_2}}$ | Diferencia de Proporciones |
| $\sigma^2$ | $\frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{\alpha/2}}$ Ambos $\chi^2$ con cola inferior | Varianza |
| $\frac{\sigma_1^2}{\sigma_2^2}$ | $\frac{s_1^2}{s_2^2}\frac{1}{F_{1-\alpha/2,gl_1,gl_2}} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2}F_{1-\alpha/2,gl_2,gl_1}$ Ambos $F$ con cola inferior | Cociente de Varianzas |

Cuadro 1: Intervalos de confianza